

WHAT MUST BE CONSERVED: THIRTEEN CONDITIONS ON THE MISSING QUANTITY FOR THE THREE-DIMENSIONAL NAVIER–STOKES EQUATIONS

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ABSTRACT. An earlier note in this series listed eight conditions that any quantity deciding global regularity for the three-dimensional Navier–Stokes equations must satisfy. A later series of four papers, on transport, added five more: the quantity must credit advection, which every integral of a pointwise function of the velocity gradient ignores; it must not rest on invariance under volume-preserving transport alone; it must use both the exact transport of circulation and its viscous averaging; and it must control the two ways in which circulation can be delivered to a point, by gathering and by the arrival of a self-sustaining cluster of vortices. We merge the two lists and test the known candidates against all thirteen conditions. Three known results then fall into place. First, a theorem of Enciso, Peralta-Salas and Torres de Lizaaur, that helicity is the only regular integral invariant of volume-preserving transformations, shows that nothing conserved by transport alone is informative on mirror-symmetric flows. Second, we note that the heat-smoothed circulation Reynolds number at every scale is equivalent to the norm of the largest scale-critical space, $\dot{B}_{\infty,\infty}^{-1}$, where the equations are ill-posed. Third, by a theorem of Cheskidov and Shvydkoy, a solution stays regular unless that norm jumps by a fixed multiple of the viscosity, which in circulation language is the arrival at a point of a lump of circulation of order ν . What must be conserved is therefore not the circulation, which is not bounded, but its continuity in time at the critical level. This reformulates the problem and does not solve it.

1. INTRODUCTION

We write the incompressible Navier–Stokes equations on $\mathbb{R}^3$ or $\mathbb{T}^3$ as

$$\partial_t u - \nu \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \nabla \cdot u = 0, \quad (1)$$

with vorticity $\omega = \text{curl } u$ and strain $S = \frac{1}{2}(\nabla u + \nabla u^T)$. The equations are invariant under $u_\lambda(x, t) = \lambda u(\lambda x, \lambda^2 t)$. The energy is bounded a priori and is supercritical by half a derivative; no scale-critical quantity is known to be bounded a priori. This note does not close that gap.

Paper IX of this series [24] recorded eight conditions that a quantity closing the gap must satisfy, each tied to a theorem or a construction that rules out the alternative. The transport series, Papers X–XIII [25, 26, 27, 28], approached the gap from the Lagrangian side and ended with a precise description of what a singularity would have to contain. Each paper of that series excludes a further class of candidates or arguments. This note merges the two lists into one specification (§3), tests the known candidates against it (§6), and records what the merged list points to.

Three known results turn out to fit the specification closely, and we give them their own sections. The theorem of Enciso, Peralta-Salas and Torres de Lizaaur that helicity is the only integral invariant of volume-preserving transformations [8] settles in one line which conserved quantities of transport can help (§4). An elementary equivalence identifies the circulation Reynolds number at every scale, the currency of the transport series, with the norm of $\dot{B}_{\infty,\infty}^{-1}$,

the largest critical space (§5). And the regularity criterion of Cheskidov and Shvydkoy [5], which requires only that this norm not jump by a multiple of ν , then becomes a statement about the delivery of circulation to a point, the event whose anatomy was described in [28, §10]. Section 7 states the result.

The conditions are of three kinds, and we mark each: *theorems*, where a candidate violating the condition is refuted by an explicit solution or construction; *inferences* from model equations, which constrain arguments and could in principle be circumvented; and *heuristics*. Nothing here proves or disproves regularity.

2. THE SETTING

As in [24], a functional Φ on smooth divergence-free fields *decides regularity* if (i) along every smooth solution on $[0, T)$, $\sup_{t < T} \Phi(u(t)) \leq F(\Phi(u_0), \|u_0\|_{L^2})$ with F independent of T ; and (ii) if $\sup_{t < T} \Phi(u(t)) < \infty$ then u extends smoothly past T . Energy satisfies (i) and not (ii); every critical norm satisfies (ii) and is not known to satisfy (i).

Circulation and viscosity have the same dimension, $\text{length}^2/\text{time}$. For a closed curve C the circulation $\Gamma(C) = \oint_C u \cdot dr$ satisfies $\Gamma_{u_\lambda}(C) = \Gamma_u(\lambda C)$, so the *circulation Reynolds number* Γ/ν is dimensionless and scale-invariant. It is the only such number that can be formed from a curve, and it is the currency of [28]: circulation is never created by the dynamics, only carried, compressed from larger scales, and lost to the viscous average [28, Prop. 5.1, Cor. 5.2].

3. THIRTEEN CONDITIONS

The eight conditions of Paper IX. We restate them briefly; the justifications are in [24, §§3–7].

Condition C1 (scaling). Φ is invariant under $u \mapsto u_\lambda$, or controls a quantity that is.

Condition C2 (a priori). The bound (i) holds for all smooth data.

Condition C3 (coercivity). Φ controls a critical norm strongly enough for (ii) to follow.

Condition C4 (structure). The proof of (i) does not apply to Tao’s averaged equation [17], nor to the Euler–Arnold equations with inhomogeneous inertia operators near the Euler metric [18]. *Theorem*.

Both constructions blow up. The first shares the energy identity and the harmonic-analysis estimates of (1); its cascade operator is not of Euler–Arnold form for any positive-definite inertia operator, which is what Papers I–IV establish [19, 20, 21, 22], so it has no Kelvin theorem. The second has the Lie–Poisson structure and a Kelvin theorem for its momentum, but no viscosity.

Condition C5 (viscosity). If Φ is also bounded along Euler solutions, the passage (ii) uses $\nu > 0$ essentially [7, 4]. *Theorem*.

Condition C6 (limits). Φ is lower semicontinuous under local convergence and invariant under translation and scaling, so that its bound passes to blowup limits and to the bounded ancient solutions of Koch, Nadirashvili, Seregin and Šverák [10].

Condition C7 (parity). Φ is informative on the class of mirror-symmetric solutions, which (1) preserves. *Theorem*: every pseudoscalar vanishes on that class [24, Prop. 4.1], and the class contains the known blowup ansatzes for model equations [13].

Condition C8 (transport). The proof of (i) uses the advection term. *Inference* from Miller’s family of strain models [15, 14], which share the constraint space, the scaling and the enstrophy identity of (1), are regular at one end and blow up in the interior [24, §7].

Five conditions from the transport series.

Condition C9 (non-locality). Φ is not an integral of a pointwise function of the velocity gradient, nor controlled only through such integrals. It is nonlocal (two-point, loop or flow-map based) or of maximum type. *Theorem*, given C8.

For divergence-free u and every C^1 function F of the gradient, $\int u \cdot \nabla F(\nabla u) dx = 0$ [25, Prop. 5.1]. Advection therefore contributes nothing to $\frac{d}{dt} \int F(\nabla u)$ for any F , at every Lebesgue exponent, and an argument that satisfies C8 cannot be run on such integrals alone. The quantities that do see transport are those that are carried by it: maxima of transported scalars, and quantities attached to fluid labels, curves or surfaces. On the flow-map side, Paper XI shows that the two natural candidates fail: the L^2 geometry of the volume-preserving group is blind to regularity (Shnirelman’s finite diameter, [26, Thm. 4.1]), and the stretching functional of Crippa and De Lellis, extended to the stochastic flow, has exactly the scaling of the energy [26, Thm. 5.2].

Condition C10 (beyond transport invariants). The bound (i) does not come from invariance under volume-preserving transport alone. *Theorem*; see §4.

Condition C11 (circulation and averaging). The proof of (i) uses both the exact transport of circulation (a Kelvin theorem, pathwise in the stochastic Lagrangian representation of Constantin and Iyer [6]) and its viscous averaging. *Inference*.

By [28, Prop. 5.1], for every closed curve C and $s \leq t$, $\oint_C u(t) \cdot dr = \mathbb{E} \oint_{A_{s,t}(C)} u(s) \cdot dr$, where $A_{s,t}$ is the back-to-labels map of the stochastic flow. Equation (1) has both ingredients. Every blowup construction for a neighbouring equation that we know of gives up one of them [28, §5]: the Euler constructions [7, 4, 18, 16] keep a Kelvin theorem and have no averaging; Tao’s averaged equation and Miller’s strain models keep viscosity and have no circulation theorem. The inference is that a regularity mechanism, if there is one, lives in the combination. It is an inference from the absence of a counterexample, not a theorem.

Condition C12 (gathering). Φ bounds a priori the circulation Reynolds number that can be collected into small curves. *Theorem* that it is needed, in the sense of Remark 3.1.

Circulation is not created, but it can be gathered from larger curves. The only bound we have, $\Gamma_t^{\text{cap}}(\ell) \leq \Gamma_s^{\text{cap}}(\ell e^{\Lambda_{s,t}})$ with $\Lambda_{s,t} = \int_s^t \|\nabla u\|_{L^\infty}$ [28, Cor. 5.2], gives nothing at a singularity, where $\Lambda \rightarrow \infty$. With the heuristic energy $\Gamma^2 \ell$ of a structure of circulation Γ and size ℓ , the energy allows circulation of order $(E_0/\ell)^{1/2}$ at scale ℓ : the half-derivative gap, in the currency of circulation. In axisymmetric flow the swirl circulation ru^θ obeys a maximum principle and gathering is impossible for it; that is the scalar on which the axisymmetric Liouville theorems rest [10].

Condition C13 (arrival). Φ excludes the delivery to a point of circulation Reynolds number above an absolute constant, including at bounded circulation Reynolds number. *Theorem* that it suffices, in the form of Corollary 5.3; the identification with clusters is a heuristic.

In the exact reduction of a vortex tube in an external strain, collapse at any rate is possible, and a Navier–Stokes realization needs suppliers of total circulation Reynolds number at least 8π within a few core radii [28, Prop. 10.1, Cor. 10.2]. At bounded circulation Reynolds numbers the suppliers intensify with the core, so a collapse is a cluster of vortices that stretch one another axially, assembled with its circulation in place. A Type I collapse is a bounded trajectory of Leray’s equation in similarity variables, and a Type II collapse below the parabolic scale rescales to a bounded eternal solution of (1) [28, Rem. 10.3]. Bounding the circulation Reynolds number, which is C12, does not exclude either.

Remark 3.1. By Corollary 5.3 below, a singularity requires the critical norm $\dot{B}_{\infty,\infty}^{-1}$ to jump by at least $c\nu$. A jump can happen in two ways: the norm grows without bound near the singular point, which by Proposition 5.1 means that circulation is gathered (C12); or it stays bounded and a lump of circulation of order ν concentrates to a point (C13). These are the two kinds of Type II collapse of [28, Rem. 10.3], and the second also covers Type I.

4. NOTHING CONSERVED BY TRANSPORT ALONE IS INFORMATIVE

Let M be a closed three-manifold with a volume form, and consider functionals on exact divergence-free vector fields ξ (on $\mathbb{T}^3$: divergence-free fields of mean zero, for instance $\xi = \omega$) that are invariant under the action $\xi \mapsto \phi_*\xi$ of volume-preserving diffeomorphisms, which is how the vorticity is carried by the Euler equations.

Theorem 4.1 (Enciso, Peralta-Salas and Torres de Lizaur [8]). *Every regular integral invariant of volume-preserving transformations on exact divergence-free fields is a function of the helicity.*

Corollary 4.2. *Every regular integral invariant of volume-preserving transformations is constant on the mirror-symmetric class of C7. In particular, no such invariant satisfies C7.*

Proof. By Theorem 4.1 it equals $f(H)$, and $H = 0$ on the mirror-symmetric class [24, Prop. 4.1]. $\square$

This is the content of C10, and it explains the dimensional split of the programme. In two dimensions the vorticity is a scalar and is carried as a function; every $\int f(\omega)$ is conserved by the Euler flow, the ones with f even are parity-even, and they close the problem, for the two-dimensional Euler equations and for the whole H^s Euler–Arnold family, $s > 0$ [23, Prop. 7.1]. In three dimensions the vorticity is carried as a field whose lines are stretched, the family of such invariants collapses to the helicity, and the helicity is a pseudoscalar. The axisymmetric scalar $\Gamma = ru^\theta$ escapes Corollary 4.2 only because it is conserved by a restricted class of transformations, those commuting with the symmetry: it is Kelvin’s theorem on the circles around the axis, the one family of curves that an axisymmetric flow maps to itself.

So the quantity sought cannot be conserved by transport. Whatever bound it satisfies must come from the viscous average or from the specific Biot–Savart coupling, which agrees with C5 and C11.

5. CIRCULATION AT EVERY SCALE IS THE LARGEST CRITICAL SPACE

For $t > 0$ let $e^{t\Delta}$ be the heat semigroup, with $\sqrt{t}$ a length. For a divergence-free u on $\mathbb{R}^3$ in the space of tempered distributions modulo polynomials for which $e^{t\Delta}u \rightarrow 0$ as $t \rightarrow \infty$, or of mean zero on $\mathbb{T}^3$, write

$$\|u\|_{\dot{B}_{\infty,\infty}^{-1}} = \sup_{t>0} \sqrt{t} \|e^{t\Delta}u\|_{L^\infty},$$

which is the heat characterization of the homogeneous Besov norm [2, Thm. 2.34]. Fix $\varepsilon_0 > 0$, chosen in the proof below, and for $t > 0$ let $\mathcal{C}_t$ be the set of oriented circles of radius $\varepsilon_0\sqrt{t}$, with any centre and in any plane. Define the *smoothed circulation Reynolds number*

$$R^\sharp(u) = \frac{1}{\nu} \sup_{t>0} \sup_{C \in \mathcal{C}_t} \left| \oint_C e^{t\Delta}u \cdot dr \right|,$$

the largest circulation Reynolds number of u seen at any scale $\sqrt{t}$ after smoothing at that scale.

Proposition 5.1. *There are absolute constants $\varepsilon_0, c_1, c_2 > 0$ such that*

$$c_1 \|u\|_{\dot{B}_{\infty,\infty}^{-1}} \leq \nu R^\sharp(u) \leq c_2 \|u\|_{\dot{B}_{\infty,\infty}^{-1}}.$$

Moreover, $|\oint_C e^{t\Delta} u \cdot dr| \leq L t^{-1/2} \|u\|_{\dot{B}_{\infty,\infty}^{-1}}$ for every closed curve C of length L .

Proof. The second statement is $|\oint_C e^{t\Delta} u \cdot dr| \leq L \|e^{t\Delta} u\|_{L^\infty}$, and with $L = 2\pi\varepsilon_0\sqrt{t}$ it gives the upper bound with $c_2 = 2\pi\varepsilon_0$.

For the lower bound let $f_t = e^{t\Delta} \omega = \operatorname{curl} e^{t\Delta} u$ and $M = \sup_{t>0} t \|f_t\|_{L^\infty}$. Writing $f_t = \operatorname{curl} e^{(t/2)\Delta} (e^{(t/2)\Delta} u)$ and using $\|\nabla e^{s\Delta} g\|_{L^\infty} \leq c_0 s^{-1/2} \|g\|_{L^\infty}$ shows $M \leq 2c_0 \|u\|_{\dot{B}_{\infty,\infty}^{-1}} < \infty$. Let $C \in \mathcal{C}_t$ bound the flat disk D of radius $\rho = \varepsilon_0\sqrt{t}$, unit normal n and centre x . By Stokes' theorem $\oint_C e^{t\Delta} u \cdot dr = \int_D f_t \cdot n$, and

$$\left| f_t(x) \cdot n - \frac{1}{\pi\rho^2} \int_D f_t \cdot n \right| \leq \rho \|\nabla f_t\|_{L^\infty} \leq \rho c_0 (t/2)^{-1/2} \|f_{t/2}\|_{L^\infty}.$$

Multiplying by t , and using $\frac{t}{2} \|f_{t/2}\|_{L^\infty} \leq M$,

$$t |f_t(x) \cdot n| \leq \frac{\nu R^\sharp(u)}{\pi\varepsilon_0^2} + 2\sqrt{2} c_0 \varepsilon_0 M.$$

Taking the supremum over x , n and t and choosing $\varepsilon_0 = 1/(4\sqrt{2}c_0)$ gives $M \leq 2\nu R^\sharp(u)/(\pi\varepsilon_0^2)$. Finally M is the heat characterization of $\|\omega\|_{\dot{B}_{\infty,\infty}^{-2}}$, and for divergence-free u the operator $\operatorname{curl}(-\Delta)^{-1}$, a Fourier multiplier homogeneous of degree -1 and smooth away from the origin, maps $\dot{B}_{\infty,\infty}^{-2}$ boundedly to $\dot{B}_{\infty,\infty}^{-1}$ [2, Ch. 2]. Hence $\|u\|_{\dot{B}_{\infty,\infty}^{-1}} \leq c M \leq c_1^{-1} \nu R^\sharp(u)$. $\square$

The proposition is elementary, and we state it because of what it identifies. The circulation Reynolds number at every scale, which the transport series reached from the Lagrangian side as the currency of the ledger, is, after smoothing at the scale of observation, the norm of $\dot{B}_{\infty,\infty}^{-1}$. Every translation-invariant Banach space of distributions with the scaling of (1) embeds continuously into $\dot{B}_{\infty,\infty}^{-1}$, so it is the largest critical space [12, 3]. It is also where the well-posedness theory ends: Koch and Tataru proved global well-posedness for small data in BMO^{-1} [11], and Bourgain and Pavlović proved norm inflation, hence ill-posedness, in $\dot{B}_{\infty,\infty}^{-1}$ [3]. A bound on $R^\sharp$ alone is therefore not expected to satisfy C3. That agrees with the anatomy: collapse at bounded circulation Reynolds number is not excluded by any statement proved in [28].

What does suffice at this level is continuity.

Theorem 5.2 (Cheskidov and Shvydkoy [5]). *There is an absolute constant $c > 0$ such that a Leray–Hopf solution of (1) which is continuous in time with values in $B_{\infty,\infty}^{-1}$ on $(0, T]$, or whose jumps in the $B_{\infty,\infty}^{-1}$ norm do not exceed $c\nu$, is regular on $(0, T]$.*

Corollary 5.3. *Let u be a smooth solution on $\mathbb{T}^3 \times [0, T)$ that is singular at T , continued as a Leray–Hopf solution. Then u has at T a jump, in the sense of [5], of size at least $c\nu$, and along a sequence $t \rightarrow T^-$ the smoothed circulation Reynolds number of the change satisfies $R^\sharp(u(t) - u(T)) \geq c_1 c$: circulation Reynolds number of order one is delivered to arbitrarily small scales.*

Proof. The solution is continuous in $B_{\infty,\infty}^{-1}$ on $(0, T)$, where it is smooth, so by Theorem 5.2 its jump at T exceeds $c\nu$. On $\mathbb{T}^3$ with mean zero the homogeneous and inhomogeneous norms are equivalent, and Proposition 5.1 converts the jump into the bound on $R^\sharp$. $\square$

The corollary is the Eulerian form of the anatomy of [28, §10]. There, a collapsing core carries a fixed circulation Γ into a radius $a(t) \rightarrow 0$, and it needs suppliers of total circulation Reynolds number at least 8π at distances comparable to a . The change $u(t) - u(T)$ then contains the core, whose smoothed circulation at the scale $a(t)$ is a fixed fraction of Γ . That is a jump of

$R^\sharp$ of order Γ/ν . The two thresholds, the constant c of Theorem 5.2 and the 8π of the tube reduction, are not comparable in any precise sense; we record only that both say the same thing, that a singularity is the arrival at a point of circulation of order ν or more.

6. THE CANDIDATES AGAINST THE LIST

	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13
Energy $\ u\ _{L^2}$	n	y	n	n	–	n	y	n	n	y	n	n	n
Helicity H	y	p	n	y	y	y	n	n	n	n	n	n	n
λ_2^+ , $\det S$	y	n	y	?	y	y	y	n	n	y	n	n	?
Swirl $\Gamma = ru^\theta$	y	s	p	–	y	y	y	y	y	y	y	s	n
Head pressure Q	y	s	n	–	y	y	y	y	y	y	n	–	?
Sparseness scale [9]	y	e	y	?	y	y	y	n	y	y	n	n	?
$R^\sharp \simeq \ u\ _{\dot{B}_{\infty,\infty}^{-1}}/\nu$	y	?	p	y	y	y	y	y	y	y	y	?	n

TABLE 1. Candidates and the thirteen conditions. y: satisfied; n: violated; p: partial; s: only in a special class (axisymmetric for Γ , stationary for Q); e: only at the level of the energy [1]; ?: not determined; –: not applicable. Columns C1–C8 for the first six rows are those of [24, Table 1]. For $R^\sharp$, C3 is partial because boundedness is the open endpoint and continuity suffices (Theorem 5.2); C2 is the open condition C12.

Table 1 extends the table of [24]. Two rows need comment. The swirl Γ satisfies the most conditions of any quantity known, but only in the axisymmetric class, and even there a bound on it does not exclude collapse without further decay [10]. The smoothed circulation Reynolds number $R^\sharp$ satisfies every structural condition, C1 and C6–C11, because it is built from the circulation, which is carried exactly and averaged; its failures are exactly the two conditions that the transport series left open, C12 and C13, together with the open condition C2. It is, in this sense, the candidate the whole list points to, and the table says precisely what is missing from it.

7. WHAT MUST BE CONSERVED

The merged list admits a short reading. By Corollary 4.2 the quantity is not conserved by transport; by C9 it is not an integral of a pointwise function; by C7 it is a true scalar; by C11 it is built from the carried and averaged circulation; by Proposition 5.1 the natural such quantity lives at the level of the largest critical space; and by Corollary 5.3 what is needed there is not a bound but continuity. What must be conserved is not the circulation, which Kelvin’s theorem carries curve by curve without bounding, but the continuity in time of the circulation Reynolds number at every scale.

Question 7.1. *Is there an absolute constant $c > 0$ such that no smooth solution of (1) delivers circulation of $c\nu$ or more to a point? Equivalently, by Proposition 5.1 and Theorem 5.2, can the viscous average of the stochastic Kelvin theorem be shown to shave any circulation carried to scales $\ell \rightarrow 0$ below $c\nu$ before it arrives?*

The second form is Question 5.3 of [28], the circulation transport question, at its critical threshold, and it is also the question of Paper IX, Question 9.1, on a parity-even, scale-invariant quantity whose control uses the transport term, made specific. We stress that Question 7.1 is equivalent to the regularity problem by Theorem 5.2, so it is a reformulation and not a reduction

in difficulty. What the specification adds is that three independent approaches meet at one object: the Eulerian theory of critical spaces, which ends at $\dot{B}_{\infty,\infty}^{-1}$; the Lagrangian theory of circulation, which ends at the circulation Reynolds number; and the geometric theory of invariants, which rules out every conserved quantity except one that parity makes useless.

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